Lakshman Rao Botuku

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PROFESSIONAL SUMMARY

AI & Backend Developer with ~1 year of professional experience in Python, Django REST Framework, FastAPI, and Machine Learning. Hands-on experience in building LLM-based applications, Retrieval-Augmented Generation (RAG) systems, REST APIs, and scalable backend services. Strong understanding of NLP, vector search, authentication, and system design. Passionate about developing production-ready AI solutions.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript

AI / Machine Learning & NLP: TensorFlow, HuggingFace Transformers, OpenAI GPT (basic usage)

LLM & RAG Frameworks: LangChain, LlamaIndex (exposure), Prompt Engineering

Backend & APIs: Fast API, Django, Django REST Framework, RESTful APIs, JSON, Postman

Frontend: React.js, Tailwind CSS

Databases & Vector Stores: MySQL, PostgreSQL, MongoDB, Firebase

Tools & Platforms: Git, GitHub, Linux, VS Code, PyCharm, Jupyter Notebook, Google Colab.

Development Methodologies: MVT Architecture, Responsive Web Design, OOP Principles

Core Concepts: AI Pipelines, Model Inference, Semantic Search, System Design, Data Processing, Scalable Architecture

WORK EXPERIENCE

Quantum Works Private Limited — Software Developer (Trainee → Full-Time) Mar 2025 - Present

Key Projects

Enterprise Document Question Answering System (LLM + RAG) - Sept 2025 – Jan 2026

- Designed and developed an **end-to-end Retrieval-Augmented Generation (RAG) system** to enable accurate question answering from enterprise-scale documents.
- Built **document ingestion and preprocessing pipelines**, including chunking and embedding generation using **LangChain and HuggingFace**, improving retrieval efficiency.
- Implemented **vector search using FAISS/Chroma**, enabling fast semantic retrieval across large document datasets.
- Integrated multiple LLMs (**OpenAI GPT, LLaMA, Mistral**) to generate **context-aware responses**, improving answer relevance and reducing incorrect outputs.
- Developed **FastAPI-based backend services** for real-time inference, supporting scalable and low-latency API responses.
- Built REST APIs handling **100+ requests/min with optimized response time and efficient resource utilization**.
- Implemented **metadata filtering and relevance ranking**, improving retrieval precision and reducing hallucination in generated responses.

Wound Detection & Healing Analysis System

- Developed an **AI-powered wound detection and classification system** to analyze medical images and assist in clinical decision-making.
- Implemented advanced image preprocessing and feature extraction techniques using **OpenCV**, improving input data quality for model training.
- Built and trained **machine learning models** to classify wound types and estimate healing progression with improved prediction accuracy.
- Designed and developed **FastAPI-based RESTful services** for image upload and real-time result delivery, enabling seamless integration with frontend applications.

- Integrated backend services for **real-time wound analysis**, ensuring low-latency predictions and efficient processing.
- Implemented **healing trend analysis** to monitor patient recovery over time, supporting data-driven insights for treatment evaluation.
- Developed secure **authentication and role-based access control (RBAC)** for patients and healthcare providers, ensuring data privacy and access control.
- Managed patient data and medical reports using **structured database systems**, enabling efficient storage, retrieval, and reporting.
- Improved model performance through **data preprocessing, validation, and hyperparameter tuning**, enhancing prediction reliability.
- Designed a **scalable and modular backend architecture**, ensuring production readiness and future extensibility.

EDUCATION

Narasaropeta Engineering College
Bachelor of Technology CSE-(Artificial Intelligence) ***Aug 2020 – May 2024***

Projects

Labour Management System – Labour Availability & Management System

- Designed and developed a **Labour Availability & Management System** to efficiently manage workers, contractors, and job assignments in a centralized platform.
- Built scalable backend services using **Django REST Framework**, enabling functionalities such as worker registration, job posting, attendance tracking, and wage management.
- Implemented **role-based authentication and authorization (RBAC)** for Admin, Contractor, and Labour users, ensuring secure and controlled access.
- Integrated **PostgreSQL** for reliable transactional data storage, supporting efficient querying and reporting of workforce data.
- Designed a **scalable and modular system architecture**, enabling future integration of AI-based features such as worker recommendation and demand forecasting.
- Developed clean and well-structured **RESTful APIs**, following microservices-oriented and modular design principles for maintainability and scalability.